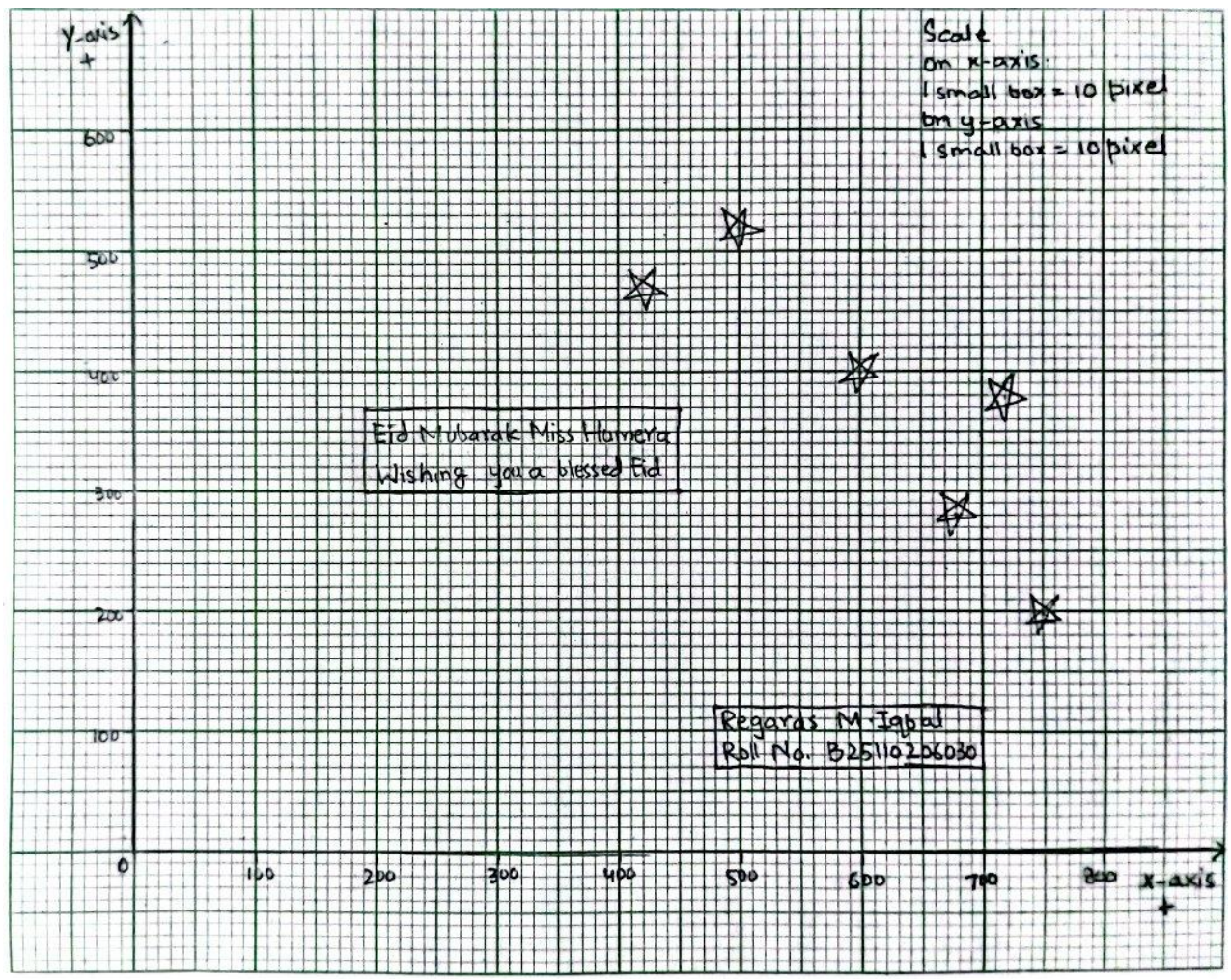
Date: 18-03-2026

Mini Project

Task: Eid Greeting Card

- ✓ **Design of card on graph sheet**
- ✓ **Flow of program**
- ✓ **Numbering of program**
- ✓ **Flowchart**
- ✓ **Logic Building**
- ✓ **Program with labeling**
- ✓ **Output**

1. Design of Card on Graph Sheet



2. Flow of Program

1. **Initialization:** The program imports necessary libraries (OpenGL, Windows Multimedia, RGBA) and defines the global window dimensions (800x600).
2. **Window Creation:** The main() function initializes GLUT, sets the display mode to RGB, and creates the application window.

3. **Environment Setup:** The myInit() function configures a 2D orthographic projection matrix so that OpenGL coordinates match standard pixel coordinates.
4. **Asset Loading:** Within myInit(), the 24-bit .bmp background image is loaded into memory, and the .wav audio file is triggered to play asynchronously on a loop.
5. **Render Trigger:** glutDisplayFunc() calls the display() function to begin rendering graphics.
6. **Layer 1 (Background):** The screen buffer is cleared, the raster position is set to the origin (0,0), and the image pixel data is drawn to the screen.
7. **Layer 2 (Geometry):** The drawStar() function is called multiple times, executing trigonometric loops to draw GL_POLYGON stars at specific coordinates.
8. **Layer 3 (Typography):** The renderText() function iterates through character arrays, rendering the greeting and student details using GLUT bitmap fonts.
9. **Execution:** glFlush() forces the rendered layers to the screen, and the program enters glutMainLoop().

3. Flowchart

```
 [ Start Program ]
      |
      v
[ Initialize GLUT & Create 800x600 Window ]
      |
      v
[ Execute myInit() Function ]
      |-- Set gluOrtho2D(0, 800, 0, 600)
      |-- Load "eid_bg.bmp" via RGBApixmap
      |-- Trigger PlaySound() on separate thread
      v
[ Enter glutMainLoop() & Call display() ]
      |
      v
[ Clear GL_COLOR_BUFFER_BIT ]
      |
      v
[ Draw BMP Background at Origin (0,0) ]
      |
      v
[ Generate GL_POLYGON Stars via Trigonometry ]
      |
      v
[ Render Bitmap Fonts for Greetings & Details ]
      |
      v
[ Flush Buffer to Screen (glFlush) ]
```

4. Logic Building

- **Coordinate System Logic:** By using gluOrtho2D(0, screenWidth, 0, screenHeight), the program maps the OpenGL coordinate system to a precise 1:1 pixel grid. This makes placing UI elements, like text and stars, mathematically straightforward rather than relying on normalized device coordinates (-1.0 to 1.0).
- **Geometric Generation (The Stars):** Instead of using static images for stars, the drawStar function logically generates them dynamically. It uses a for loop with 10 iterations, calculating vertices using sin() and cos(). By alternating between a large outer radius and a smaller inner radius using the modulo operator (i % 2 == 0), it creates the sharp points of a star within a single GL_POLYGON.
- **Asset Management:** The custom RGBA.h library is utilized to bypass complex OpenGL texture wrapping and filtering. It logically parses the uncompressed raw pixel data of a 24-bit BMP file and blits it directly to the frame buffer.
- **Audio Multithreading:** The logic behind the audio relies on the SND_ASYNC flag in the Windows API. This ensures the audio executes on a background thread, preventing the PlaySound function from blocking the main OpenGL rendering loop.

5. Program with Labeling and Numbering

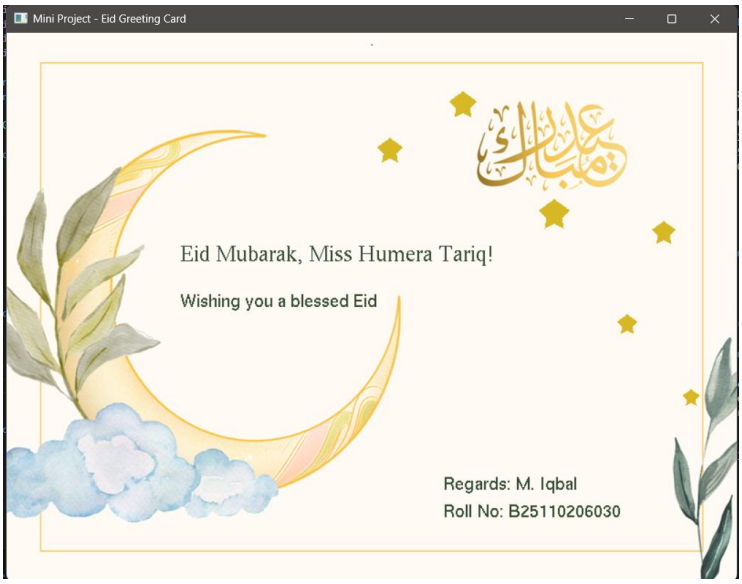
MiniProject > card.cpp > myInit()

```cpp
// [1] LIBRARIES & GLOBAL VARIABLES
#include <windows.h>
#include <GL/freeglut.h>
#include <iostream>
#include <math.h>
#include "RGBA.h"

int screenWidth = 800;
int screenHeight = 600;
RGBApixmap bgImage;

// [2] GEOMETRY LOGIC (Star Generation)
void drawStar(float x, float y, float radius) {
    glColor3f(0.85f, 0.72f, 0.15f);
    glBegin(GL_POLYGON);
    for (int i = 0; i < 10; i++) {
        float r = (i % 2 == 0) ? radius : radius / 2.0f;
        float angle = i * 3.14159f / 5.0f;
        glVertex2f(x + r * sin(angle), y + r * cos(angle));
    }
    glEnd();
}

// [3] TYPOGRAPHY LOGIC (Text Rendering)
void renderText(float x, float y, void* font, const char* string) {
    glColor3f(0.2f, 0.3f, 0.2f);
    glRasterPos2f(x, y);
    while (*string) {
        glutBitmapCharacter(font, *string++);
    }
}

// [4] ENVIRONMENT & ASSET INITIALIZATION
void myInit() {
    glClearColor(1.0f, 1.0f, 1.0f, 1.0f);

    // Set 2D Orthographic Projection
    glMatrixMode(GL_PROJECTION);
    glloadIdentity();
    gluOrtho2D(0, screenWidth, 0, screenHeight);
    glMatrixMode(GL_MODELVIEW);
    glLoadIdentity();

    // Load Image and Audio
    bgImage.readBMPFile("eid_bg.bmp", 1);
```

MiniProject > card.cpp > display()

```cpp
void myInit() {
    PlaySound(TEXT("eid_audio.wav"), NULL, SND_ASYNC | SND_FILENAME | SND_LOOP);
}

// [5] MAIN DISPLAY RENDERING PIPELINE
void display() {
    glClear(GL_COLOR_BUFFER_BIT);

    // Layer 1: Background Image
    glRasterPos2f(0, 0);
    bgImage.mDraw();

    // Layer 2: Geometric Stars
    drawStar(600, 400, 18);
    drawStar(720, 380, 14);
    drawStar(500, 520, 16);
    drawStar(680, 280, 12);
    drawStar(420, 470, 15);
    drawStar(750, 200, 10);

    // Layer 3: Text Overlays
    renderText(190, 350, GLUT_BITMAP_TIMES_ROMAN_24, "Eid Mubarak, Miss Humera Tariq!");
    renderText(190, 300, GLUT_BITMAP_HELVETICA_18, "Wishing you a blessed Eid");

    renderText(480, 100, GLUT_BITMAP_HELVETICA_18, "Regards: M. Iqbal");
    renderText(480, 70, GLUT_BITMAP_HELVETICA_18, "Roll No: B25110206030");

    glFlush();
}

// [6] MAIN EXECUTION LOOP
int main(int argc, char** argv) {
    glutInit(&argc, argv);
    glutInitDisplayMode(GLUT_SINGLE | GLUT_RGB);

    glutInitWindowSize(screenWidth, screenHeight);
    glutInitWindowPosition(100, 100);
    glutCreateWindow("Week 07 Mini Project - Eid Greeting Card");

    myInit();
    glutDisplayFunc(display);

    glutMainLoop();
    return 0;
}
```

6. Output:


Mini Project - Eid Greeting Card
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Eid Mubarak, Miss Humera Tariq!
Wishing you a blessed Eid
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